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ASSIGNMENT No: 30

TITLE: Write a JSP application to demonstrate CRUD

Problem Statement

Write a JSP application to demonstrate CRUD operations.

Learning Objectives

To implement Create, Read, Update, and Delete operations using JSP and a database.

Theory

CRUD stands for Create, Read, Update, and Delete, which represent the four basic database operations performed in web applications. JSP works together with Servlets and JDBC to implement these operations. Users interact with HTML forms created using JSP, while JDBC executes SQL statements to manipulate database records. INSERT adds new records, SELECT retrieves information, UPDATE modifies existing data, and DELETE removes unwanted records. Proper validation, exception handling, and session management improve application reliability and security. JSP CRUD applications are widely used in employee management systems, online shopping portals, hospital management systems, banking applications, and other enterprise software.

Code:

DBConnect.java

```
package E30;
import java.sql.Connection;
import java.sql.DriverManager;

public class DBConnect {
   public static Connection getConnection() throws Exception {
      Class.forName("com.mysql.cj.jdbc.Driver");
                                                                  return
DriverManager.getConnection("jdbc:mysql://localhost:3306/sit_java", "root",
"12345");
   }

}
```

Index.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>
<%@ include file="db.jsp" %>
<html>
<body>
  <h2>Item CRUD Operations</h2>
  <a href="add.jsp">Add New Item</a><br><br>
  <table border="1" cellpadding="6">
    <tr>
      <th>ID</th>
      <th>Item Name</th>
      <th>Category</th>
      <th>Price</th>
      <th>Action</th>
    </tr>
<%
  Statement st = con.createStatement();
  ResultSet rs = st.executeQuery("SELECT * FROM crud_items");
  while (rs.next()) {
    int id = rs.getInt("id");
%>
    <tr>
      <td><%= id %></td>
      <td><%= rs.getString("item_name") %></td>
      <td><%= rs.getString("category") %></td>
      <td><%= rs.getDouble("price") %></td>
      <td>
        <a href="edit.jsp?id=<%= id %>">Edit</a> |
        <a href="delete.jsp?id=<%= id %>">Delete</a>
      </td>
    </tr>
<%
  }
  con.close();
%>
  </table>
</body>
</html>
```

db.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>
<%
  Class.forName("com.mysql.cj.jdbc.Driver");
  Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/sit_java",
"root", "12345");
%>
```

add.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>
<%@ include file="db.jsp" %>
<%
 if ("POST".equals(request.getMethod())) {
   PreparedStatement ps = con.prepareStatement("INSERT INTO crud_items(item_name,
category, price) VALUES(?, ?, ?)");
   ps.setString(1, request.getParameter("item_name"));
   ps.setString(2, request.getParameter("category"));
   ps.setDouble(3, Double.parseDouble(request.getParameter("price")));
   ps.executeUpdate();
   con.close();
   response.sendRedirect("index.jsp");
   return;
 }
%>
<html>
<body>
 <h2>Add Item</h2>
 <form method="post">
   Item Name: <input type="text" name="item_name"><br><br>
   Category: <input type="text" name="category"><br><br>
   Price: <input type="text" name="price"><br><br>
   <input type="submit" value="Save">
 </form>
 <a href="index.jsp">Back</a>
</body>
</html>
```

edit.jsp

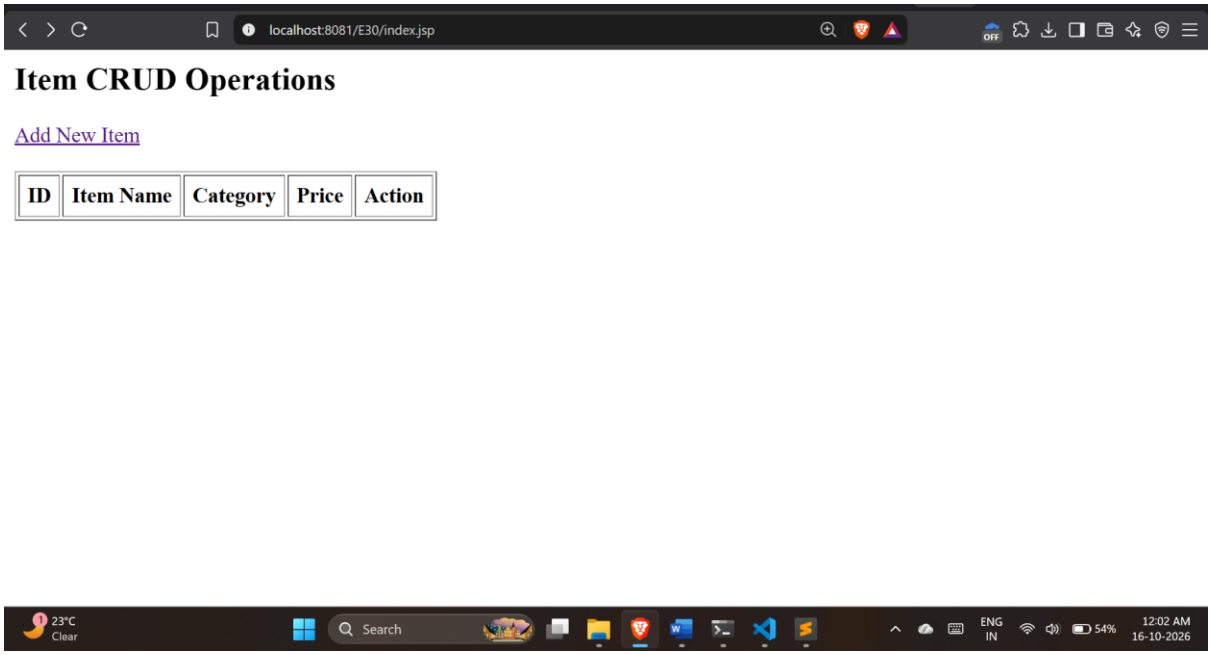
```
<%@ page import="java.lang.*, java.sql.*" %>
<%@ include file="db.jsp" %>
<%
 int id = Integer.parseInt(request.getParameter("id"));
 if ("POST".equals(request.getMethod())) {
       PreparedStatement up = con.prepareStatement("UPDATE crud_items SET
item_name=?, category=?, price=? WHERE id=?");
   up.setString(1, request.getParameter("item_name"));
   up.setString(2, request.getParameter("category"));
   up.setDouble(3, Double.parseDouble(request.getParameter("price")));
   up.setInt(4, id);
   up.executeUpdate();
   con.close();
   response.sendRedirect("index.jsp");
   return;
 }
```

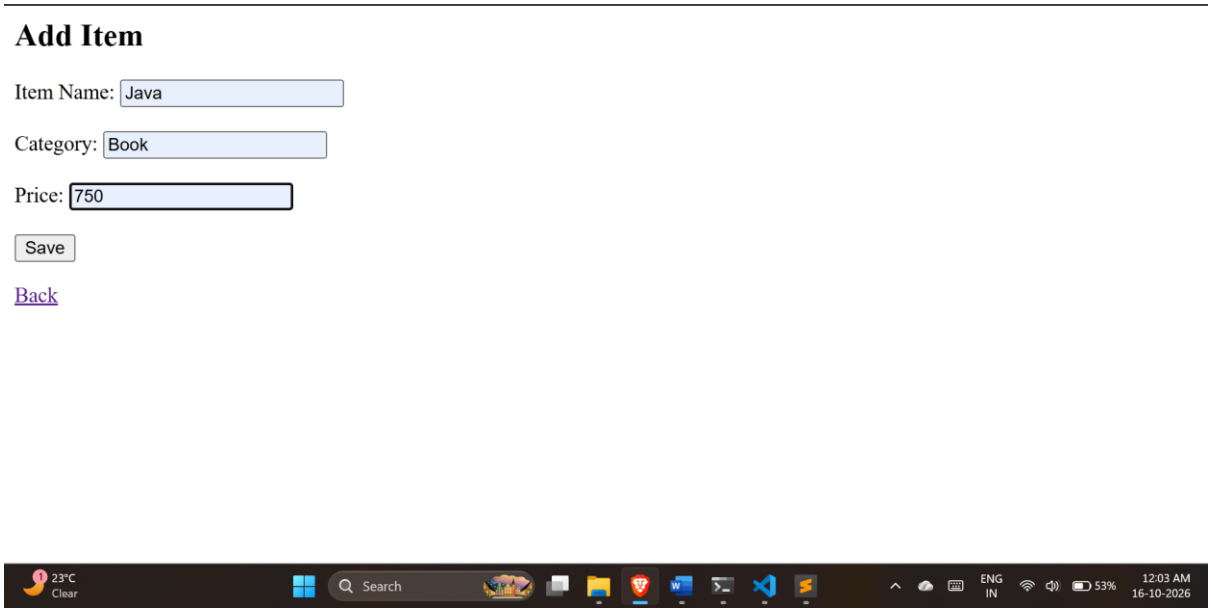
```
  PreparedStatement ps = con.prepareStatement("SELECT * FROM crud_items WHERE id=?");
  ps.setInt(1, id);
  ResultSet rs = ps.executeQuery();
  rs.next();
%>
<html>
<body>
  <h2>Update Item</h2>
  <form method="post" action="edit.jsp?id=<%= id %>">
        Item Name: <input type="text" name="item_name" value="<%= rs.getString("item_name") %>"><br><br>
    Category: <input type="text" name="category" value="<%= rs.getString("category") %>"><br><br>
      Price: <input type="text" name="price" value="<%= rs.getDouble("price") %>"><br><br>
    <input type="submit" value="Update">
  </form>
  <a href="index.jsp">Back</a>
</body>
</html>
```

delete.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>
<%@ include file="db.jsp" %>
<%
  int id = Integer.parseInt(request.getParameter("id"));
  PreparedStatement ps = con.prepareStatement("DELETE FROM crud_items WHERE id=?");
  ps.setInt(1, id);
  ps.executeUpdate();
  con.close();
  response.sendRedirect("index.jsp");
%>
```

Output:





Item CRUD Operations

Add New Item

ID	Item Name	Category	Price	Action
1	Java	Book	750.0	Edit | Delete

Update Item

Item Name: Java

Category: Book

Price: 800.0

Update

Back

Item CRUD Operations

Add New Item

ID	Item Name	Category	Price	Action
1	Java	Book	800.0	Edit | Delete

After deleted

Item CRUD Operations

Add New Item

ID	Item Name	Category	Price	Action

Exercise

1. Create a JSP-based application to add, update, delete, and view employee details.

Code:

EmployeeBean.java

```
package E30;
import java.lang.*;
import java.sql.*;

public class EmployeeBean {
   private int id;
   private String name;
   private String department;
   private double salary;
}
```

db_conn.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>
<%
  Class.forName("com.mysql.cj.jdbc.Driver");
                    Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/sit_java", "root",
"12345");
%>
```

emp_index.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>
<%@ include file="db_conn.jsp" %>
<html>
<body>
  <h2>Employee Details</h2>
  <a href="emp_add.jsp">Add New Employee</a><br><br>
  <table border="1" cellpadding="6">
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Department</th>
      <th>Salary</th>
      <th>Action</th>
    </tr>
<%
  Statement st = con.createStatement();
  ResultSet rs = st.executeQuery("SELECT * FROM emp_records");
  while (rs.next()) {
```

```
    int id = rs.getInt("id");
%>
    <tr>
      <td><%= id %></td>
      <td><%= rs.getString("name") %></td>
      <td><%= rs.getString("department") %></td>
      <td><%= rs.getDouble("salary") %></td>
      <td>
        <a href="emp_edit.jsp?id=<%= id %>">Edit</a> |
        <a href="emp_delete.jsp?id=<%= id %>">Delete</a>
      </td>
    </tr>
<%
  }
  con.close();
%>
  </table>
</body>
</html>
```

emp_add.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>

<%@ include file="db_conn.jsp" %>

<%

  if ("POST".equals(request.getMethod())) {

        PreparedStatement ps = con.prepareStatement("INSERT INTO
emp_records(name, department, salary) VALUES(?, ?, ?)");

    ps.setString(1, request.getParameter("name"));

    ps.setString(2, request.getParameter("department"));

    ps.setDouble(3, Double.parseDouble(request.getParameter("salary")));

    ps.executeUpdate();

    con.close();

    response.sendRedirect("emp_index.jsp");

    return;

  }

%>
```

```
<html>
<body>
  <h2>Add Employee</h2>
  <form method="post">
    Name: <input type="text" name="name"><br><br>
    Department: <input type="text" name="department"><br><br>
    Salary: <input type="text" name="salary"><br><br>
    <input type="submit" value="Save">
  </form>
  <a href="emp_index.jsp">Back</a>
</body>
</html>
```

emp_edit.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>
<%@ include file="db_conn.jsp" %>
<%
  int id = Integer.parseInt(request.getParameter("id"));
  if ("POST".equals(request.getMethod())) {
     PreparedStatement up = con.prepareStatement("UPDATE emp_records SET
name=?, department=?, salary=? WHERE id=?");
    up.setString(1, request.getParameter("name"));
    up.setString(2, request.getParameter("department"));
    up.setDouble(3, Double.parseDouble(request.getParameter("salary")));
    up.setInt(4, id);
    up.executeUpdate();
    con.close();
    response.sendRedirect("emp_index.jsp");
    return;
  }
  PreparedStatement ps = con.prepareStatement("SELECT * FROM emp_records
WHERE id=?");
  ps.setInt(1, id);
  ResultSet rs = ps.executeQuery();
  rs.next();
%>
<html>
```

```
<body>
  <h2>Update Employee</h2>
  <form method="post" action="emp_edit.jsp?id=<%= id %>">
     Name: <input type="text" name="name" value="<%= rs.getString("name") %>"><br><br>
          Department: <input type="text" name="department" value="<%= rs.getString("department") %>"><br><br>
     Salary: <input type="text" name="salary" value="<%= rs.getDouble("salary") %>"><br><br>
    <input type="submit" value="Update">
  </form>
  <a href="emp_index.jsp">Back</a>
</body>
</html>
```

emp_delete.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>

<%@ include file="db_conn.jsp" %>

<%

  int id = Integer.parseInt(request.getParameter("id"));

  PreparedStatement ps = con.prepareStatement("DELETE FROM emp_records WHERE id=?");

  ps.setInt(1, id);

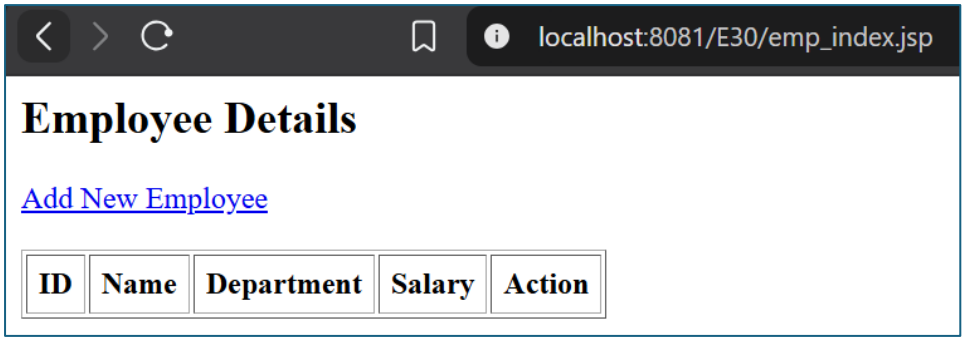
  ps.executeUpdate();

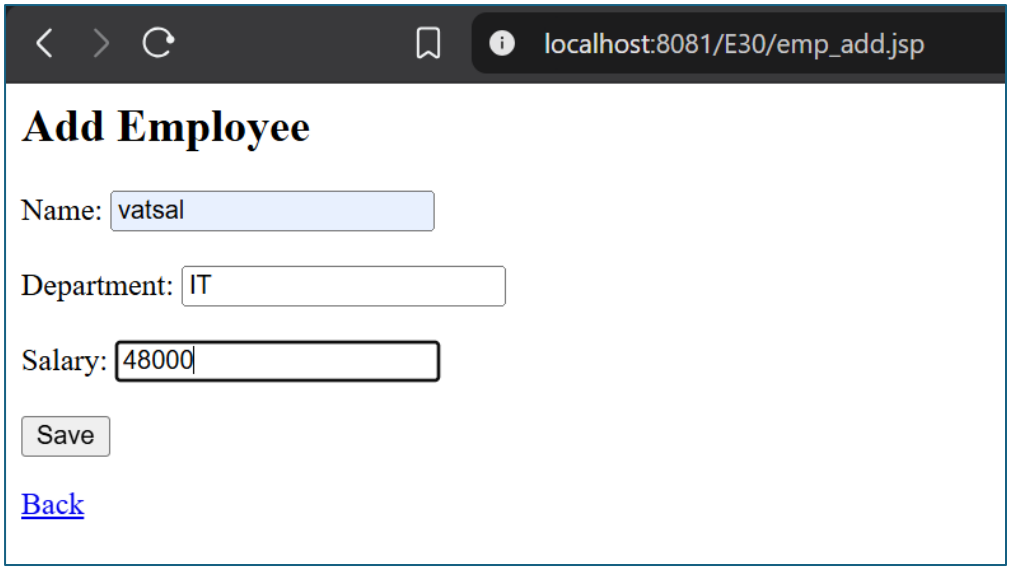
  con.close();

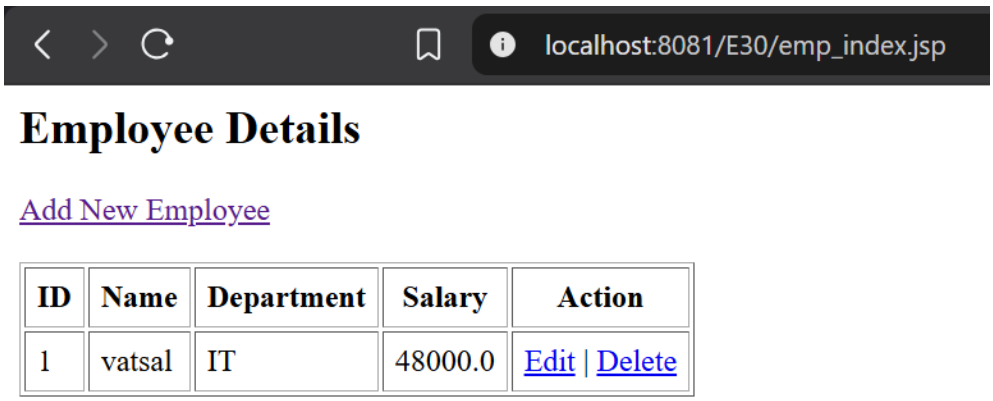
  response.sendRedirect("emp_index.jsp");

%>
```

Output:







2. Create a **JSP-based application** to manage **patient records** using **CRUD operations**. Include fields such as patient ID, name, age, disease, and treatment status.

Code:

PatientBean.java

```
package E30;
import java.lang.*;
import java.sql.*;

public class PatientBean {
   private int patientId;
   private String name;
   private int age;
   private String disease;
   private String status;

}
```

patient_index.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>
<%@ include file="db_conn.jsp" %>
<html>
<body>
  <h2>Patient Details</h2>
  <a href="patient_add.jsp">Add New Patient</a><br><br>
  <table border="1" cellpadding="6">
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Age</th>
      <th>Disease</th>
      <th>Status</th>
      <th>Action</th>
    </tr>
<%
  Statement st = con.createStatement();
  ResultSet rs = st.executeQuery("SELECT * FROM patient_records");
  while (rs.next()) {
    int id = rs.getInt("patient_id");
%>
    <tr>
      <td><%= id %></td>
      <td><%= rs.getString("name") %></td>
      <td><%= rs.getInt("age") %></td>
      <td><%= rs.getString("disease") %></td>
      <td><%= rs.getString("status") %></td>
      <td>
        <a href="patient_edit.jsp?id=<%= id %>">Edit</a> |
        <a href="patient_delete.jsp?id=<%= id %>">Delete</a>
      </td>
    </tr>
<%
  }
  con.close();
%>
  </table>
</body>
</html>
```

patient_add.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>
<%@ include file="db_conn.jsp" %>
<%
  if ("POST".equals(request.getMethod())) {
        PreparedStatement ps = con.prepareStatement("INSERT INTO
patient_records(name, age, disease, status) VALUES(?, ?, ?, ?)");
    ps.setString(1, request.getParameter("name"));
    ps.setInt(2, Integer.parseInt(request.getParameter("age")));
    ps.setString(3, request.getParameter("disease"));
    ps.setString(4, request.getParameter("status"));
    ps.executeUpdate();
    con.close();
    response.sendRedirect("patient_index.jsp");
    return;
  }
%>
<html>
<body>
  <h2>Add Patient</h2>
  <form method="post">
    Name: <input type="text" name="name"><br><br>
    Age: <input type="text" name="age"><br><br>
    Disease: <input type="text" name="disease"><br><br>
    Status: <select name="status">
      <option>Under Treatment</option>
      <option>Recovered</option>
    </select><br><br>
    <input type="submit" value="Save">
  </form>
  <a href="patient_index.jsp">Back</a>
</body>
</html>
```

patient_edit.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>
<%@ include file="db_conn.jsp" %>
<%
  int id = Integer.parseInt(request.getParameter("id"));
  if ("POST".equals(request.getMethod())) {
     PreparedStatement up = con.prepareStatement("UPDATE patient_records SET
name=?, age=?, disease=?, status=? WHERE patient_id=?");
    up.setString(1, request.getParameter("name"));
```

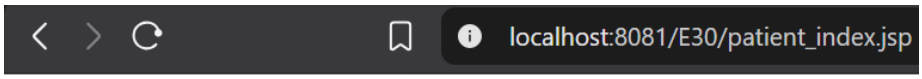
```
    up.setInt(2, Integer.parseInt(request.getParameter("age")));
    up.setString(3, request.getParameter("disease"));
    up.setString(4, request.getParameter("status"));
    up.setInt(5, id);
    up.executeUpdate();
    con.close();
    response.sendRedirect("patient_index.jsp");
    return;
  }
     PreparedStatement ps = con.prepareStatement("SELECT * FROM
patient_records WHERE patient_id=?");
  ps.setInt(1, id);
  ResultSet rs = ps.executeQuery();
  rs.next();
%>
<html>
<body>
  <h2>Update Patient</h2>
  <form method="post" action="patient_edit.jsp?id=<%= id %>">
     Name: <input type="text" name="name" value="<%= rs.getString("name″)
%>"><br><br>
        Age: <input type="text" name="age" value="<%= rs.getInt("age″)
%>"><br><br>
           Disease: <input type="text" name="disease" value="<%=
rs.getString("disease″) %>"><br><br>
     Status: <input type="text" name="status" value="<%= rs.getString("status″)
%>"><br><br>
    <input type="submit" value="Update">
  </form>
  <a href="patient_index.jsp">Back</a>
</body>
</html>
```

patient_delete.jsp

```
<%@ page import="java.lang.*, java.sql.*" %>
<%@ include file="db_conn.jsp" %>
<%
  int id = Integer.parseInt(request.getParameter("id"));
  PreparedStatement ps = con.prepareStatement("DELETE FROM patient_records
WHERE patient_id=?");
  ps.setInt(1, id);
  ps.executeUpdate();
  con.close();
```

```
  response.sendRedirect("patient_index.jsp");
%>
```

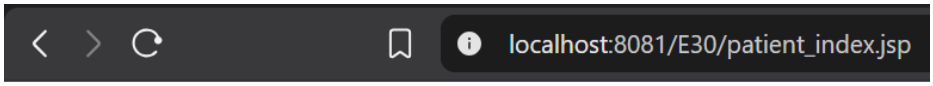
Output:



Patient Details

Add New Patient

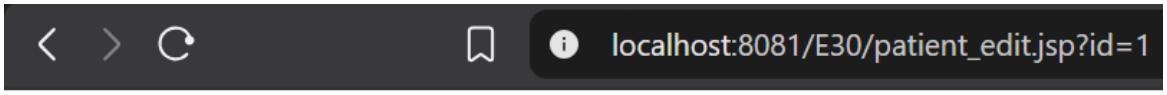
ID	Name	Age	Disease	Status	Action



Patient Details

Add New Patient

ID	Name	Age	Disease	Status	Action
1	vatsal	19	Fever	Under Treatment	Edit | Delete



Update Patient

Name: vatsal

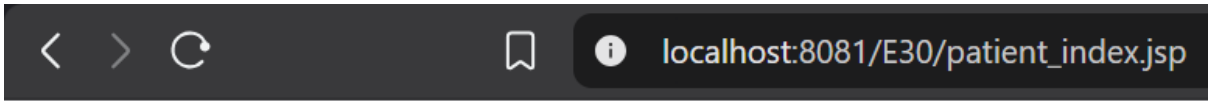
Age: 19

Disease: Fever

Status: Recoverd

Update

Back



Patient Details

Add New Patient

ID	Name	Age	Disease	Status	Action
1	vatsal	19	Fever	Recoverd	Edit | Delete

Conclusion:

I learned how to connect JSP applications to MySQL using JDBC and execute CRUD operations via SQL statements. I gained hands-on experience handling dynamic form data and manipulating database records in real time. Additionally, I understood how to manage page redirection, request parameters, and database connection lifecycles. Overall, this enhanced my practical knowledge of building data-driven web applications using Java and JSP.